

Mansoor Ahmed

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Education

- Georgia State University / Georgia Tech (Joint)**, PhD Computer Science Aug 2024 – May 2028
- Advised by Prof. Murray Patterson. GPA: 4.0/4.0.
- University of Engineering and Technology, Lahore**, B.Sc. Electrical Engineering Dec 2017 – Aug 2021
- Full scholarship recipient by Higher Education Commission Pakistan

Research Experience

- Algorithmic Biology Lab, Georgia State University** — Graduate Research Assistant Aug 2024 – present
- Advised by **Prof. Murray Patterson & Prof. Hemanth Venkateswara**.
 - Developing ML methods for antibody design and generative modeling in general.
- Lab for Understanding Network Effects (LUNE), UCLA** — Research Fellow Aug 2025 – May 2026
- Advised by **Prof. Jennifer Wilson**. Drug combination modeling for cancer subtypes via network methods.
- Data Analytics Lab, LUMS** — Research Assistant (Lahore, Pakistan) Dec 2022 – June 2024
- Advised by **Prof. Imdadullah Khan**. Graph-based methods for antigen-antibody interactions.
- Smart Data Systems Lab, LUMS** — Research Assistant (Lahore, Pakistan) Nov 2021 – Nov 2022
- Advised by **Prof. Zubair Khalid**. Software development and database management for IoT devices.

Teaching Experience

- Advanced Machine Learning (CSC 8850)** — Georgia State University Spring 2025; Fall 2025, 2026
- Graduate Teaching Assistant with **Prof. Sergey Plis**. Held office hours; created and graded problem sets and exams; delivered lectures on non-linear embedding approaches.
- Computer Org. & Programming (CS 3210)** — Georgia State University Summer 2025, Spring 2026
- Teaching Assistant. Assembly language and computer architecture.
- Spatial and Scientific Databases (CSC 8713)** — Georgia State University Fall 2024
- Mathematical Foundations for Machine Learning (EE 212)** — LUMS Fall 2022

Honors & Awards

- Molecular Basis for Disease (MBD) Fellowship, Georgia State University 2025 – present
- Warren Alpert Computational Biology and AI Network Fellow, UCLA 2025 – 2026
- Computational Genomics Summer Institute (CGSI), UCLA 2025
- EMBO Latin American Workshop on AI for Protein Design, Chile 2025
- International Summer School in Biomedical Engineering, Max Planck Institute 2023
- Higher Education Commission Undergraduate Scholarship, Pakistan 2017 – 2021

Publications

Accepted Publications

1. **ChiMERa-Bench: A Benchmark Dataset for Epitope-specific Antibody Design** 2026
M. Ahmed, N. Taj, I. U. Khan, H. Venkateswara, M. Patterson.
Transactions on ML Research (TMLR); ICLR GEM Workshop; 42+ GitHub stars
2. **Latent-Kernel Discrete Diffusion for Few-Step Generation** 2027
M. Ahmed, Y.T. Fan, H. Venkateswara, M. Patterson. *NeurIPS BeNTTo Workshop.*
3. **EvoStruct: Bridging Evolutionary and Structural Priors for Antibody CDR Design via Protein Language Model Adaptation** 2026
M. Ahmed, S. Lee, U. Khayaz, M. Patterson. *ICML GenBio Workshop.*
4. **ConTact: Contact-First Antibody CDR Design via Explicit Interface Reasoning** 2026
M. Ahmed, S. VonBank, Nadeem Taj, S. Lee, N. Jan, M. Patterson. *ICML GenBio Workshop.*
5. **Improved Graph-based Antibody-aware Epitope Prediction with PLM Embeddings** 2025
M. Ahmed, S. Ali, A. Jan, I. U. Khan, M. Patterson. *ICCABS. Springer LNBI post-proceedings.*
6. **Robust Brain Age Estimation via Regression Models and MRI-derived Features** 2023
M. Ahmed, U. Sardar, S. Ali, S. Alam, M. Patterson, I. U. Khan. *ICCCI. Springer CCIS.*

Preprints & Under Review

1. **MLD3: Mixture-of-Latents for Few-Step Discrete Diffusion Distillation** 2027
M. Ahmed, S. Tang, H. Venkateswara, M. Patterson. *Under review at ICLR.*
2. **Domino: Joint Unmasking via First-order Markov Chain for Few-Step Masked Diffusion** 2026
M. Ahmed, H. Venkateswara, M. Patterson. *Under review at TMLR.*
3. **EpiFormer: Learning Antigen-Antibody Interactions for Epitope Prediction** 2027
M. Ahmed, H. Chai, H. Wang, H. Venkateswara, M. Patterson. *Under review at ICLR.*
4. **AgForce Enables Antigen-conditioned Generative Antibody Design** 2026
M. Ahmed, M. Patterson. *Under review at AAAI (phase 2).*

Reviewing & Service

Workshops: GenBio @ ICML 2026, GEM @ NeurIPS 2026, ML4H @ NeurIPS 2026, BeNTTo @ NeurIPS 2026

Conferences: AAAI 2027, ICLR 2027, ISBRA 2025; 2026, ICCABS 2025

Journals: PeerJ Computer Science, TMLR

Organizer: GEM workshop @ ICLR 2027 (upcoming)

Skills

ML/DL: PyTorch, PyTorch Geometric, e3nn, PyTorch Lightning, Hugging Face, WandB

Protein ML: ESM-2, AntiBERTy, ProteinMPNN, AlphaFold2, OpenFold, BoltzGen, RFdiffusion

Structural Biology: BioPython, PyMOL, RosettaDock, AutoDock, BLAST, ANARCI

Programming: Python, C++, CUDA, SQL, Git, Unix/Shell, $\text{\LaTeX}$, SLURM, Docker